

Graphs of cube root functions

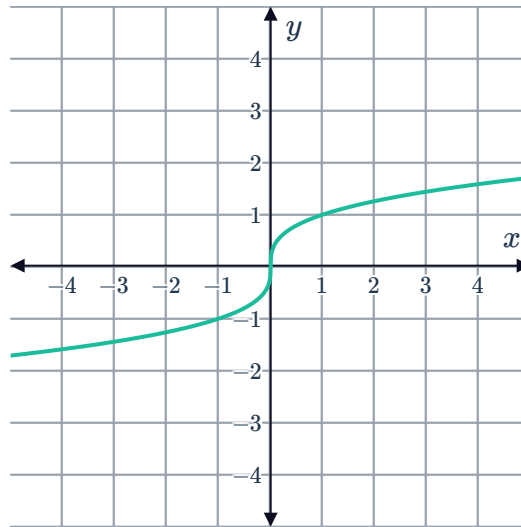


1

a

x	-100	-10	-8	-3	-1	0	1	3	8	10	100
y	-4.64	-2.15	-2	-1.44	-1	0	1	1.44	2	2.15	4.64

b



c Increasing

d No, x can be any real number.

e All real y .

f No

g No

h ∞

i As x increases, the function increases at a slower and slower rate.

j As x increases, the function increases at a faster and faster rate.

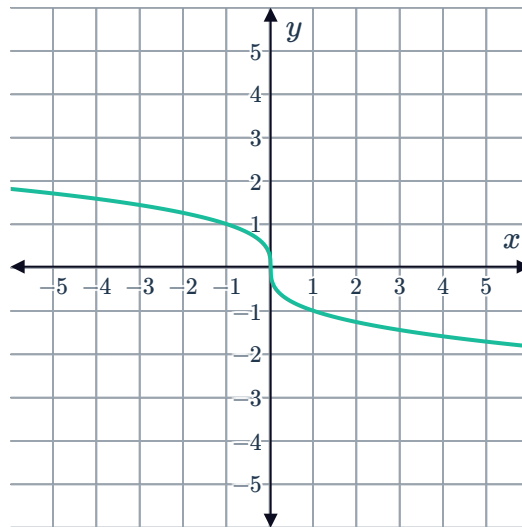
2



a

x	-100	-10	-8	-3	-1	0	1	3	8	10	100
y	4.64	2.15	2	1.44	1	0	-1	-1.44	-2	-2.15	-4.64

b



c Yes

d Yes

e All real All real x .f all real y

g Decreasing function

h As x increases, the function decreases more and more rapidly up to $x = 0$, and from $x = 0$ onwards, the rate of decrease slows down.

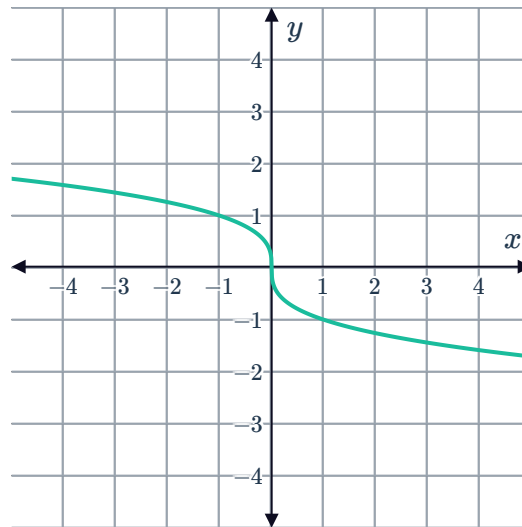
3



a

x	-100	-10	-8	-3	-1	0	1	3	8	10	100
y	4.64	-2.15	-4.64	2.15	2	1.44	1	0	-1	-1.44	-2

b



c Decreasing

d All real x e All real y f ∞ g $-\infty$

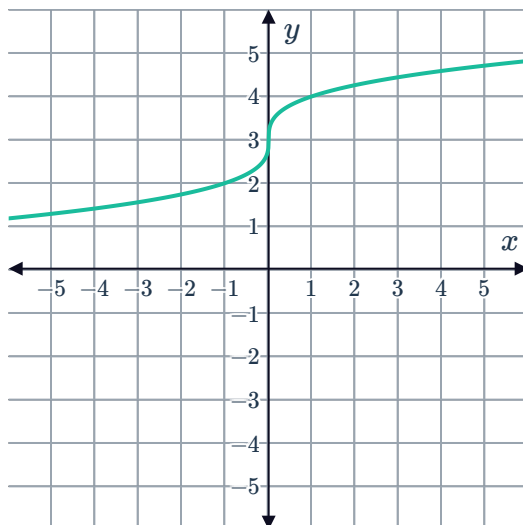
4

a Yes

b ∞ c $(0, 3)$

d 1

e



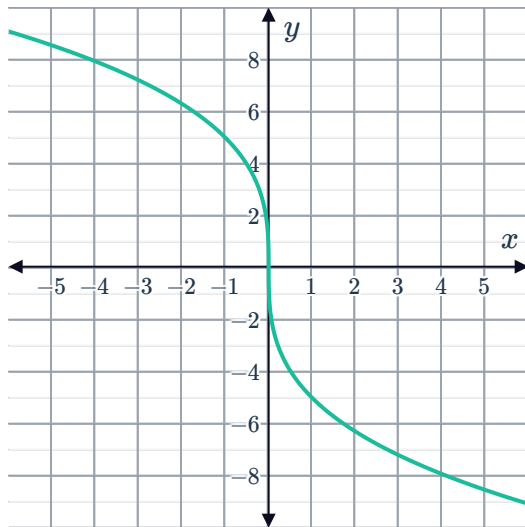


5

a All real x .

b All real y .

c

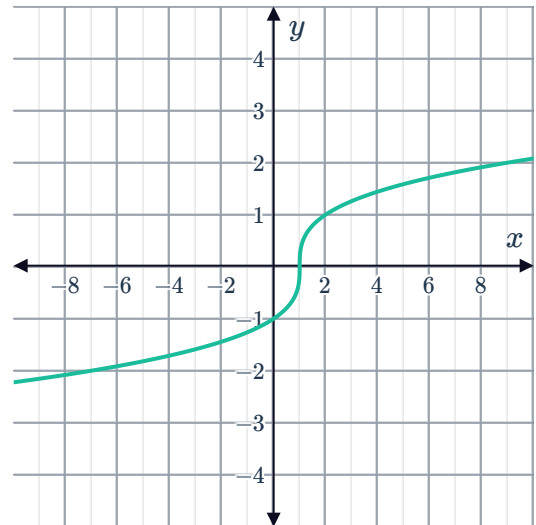


6

a

x	-7	0	1	2	9
$\sqrt[3]{x-1}$	-2	-1	0	1	2

b



c All real numbers.

d All real numbers.

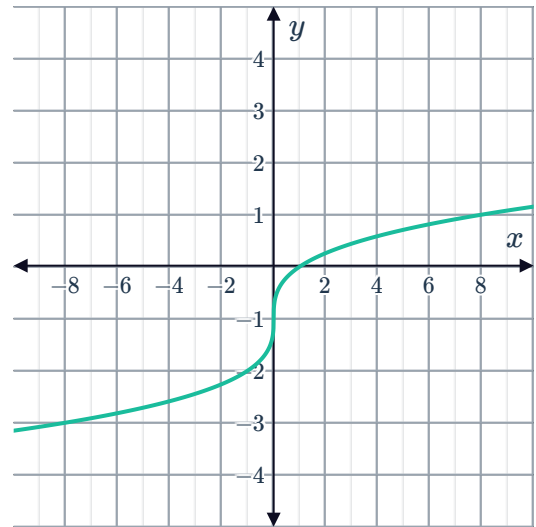
7

a

x	-8	-1	0	1	8
$\sqrt[3]{x} - 1$	-3	-2	-1	0	1



b



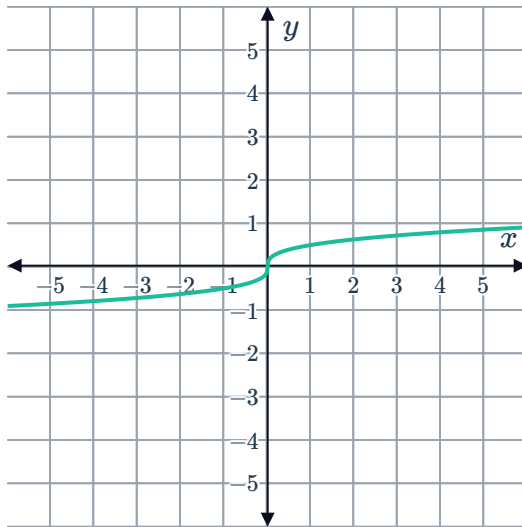
c All real numbers.

d All real numbers.

8

a

i

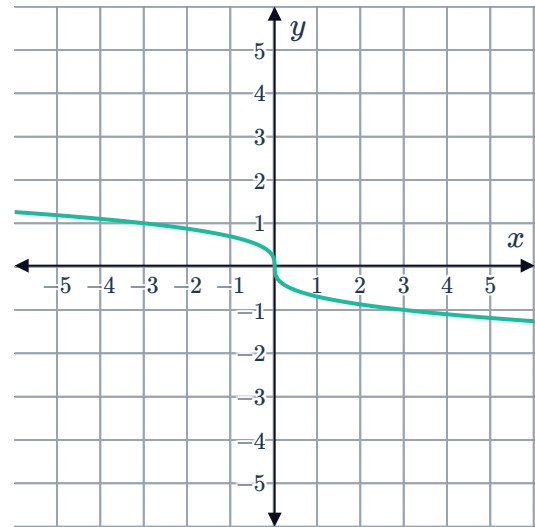


ii All real numbers.

iii All real numbers.

b

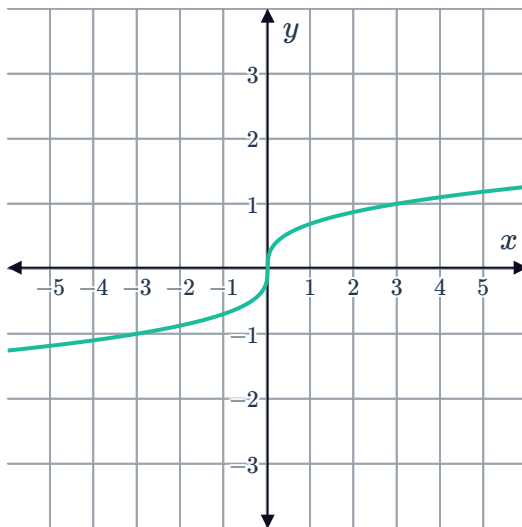
i



ii All real numbers.

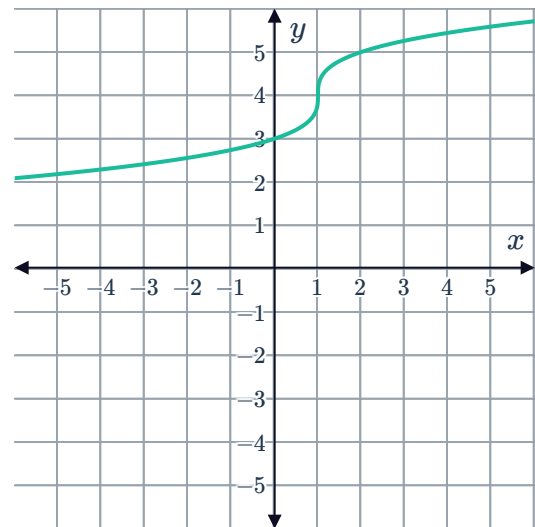
iii All real numbers.

c
i



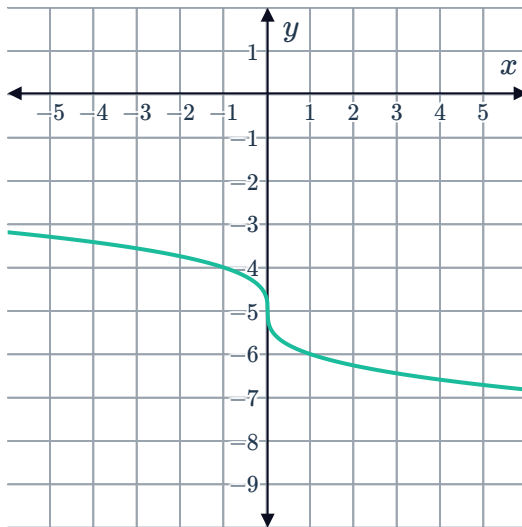
- ii All real numbers.
- iii All real numbers.

i



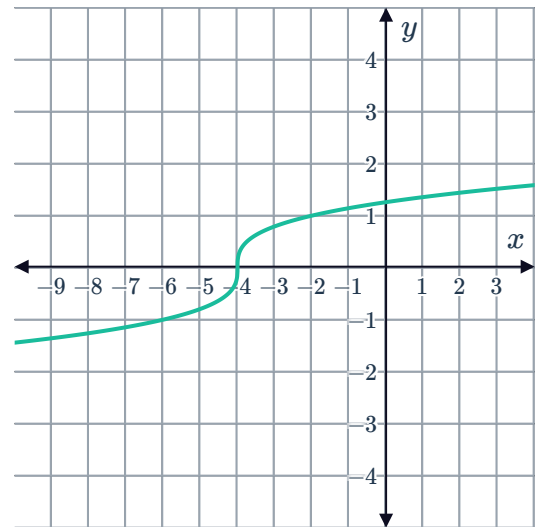
- ii All real numbers.
- iii All real numbers.

e
i



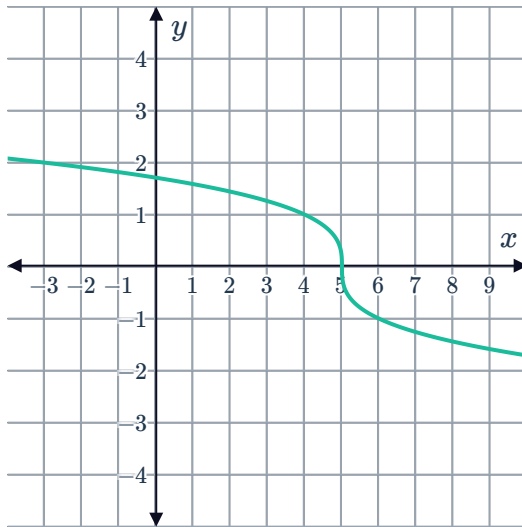
- ii All real numbers.
- iii All real numbers.

f
i



- ii All real numbers.
- iii All real numbers.

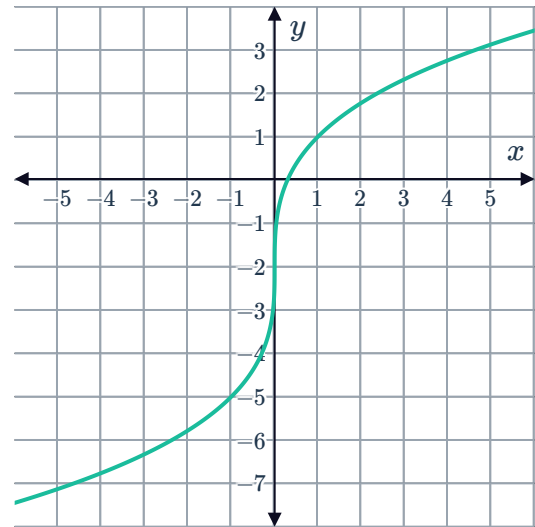
g
i



ii All real numbers.

iii All real numbers.

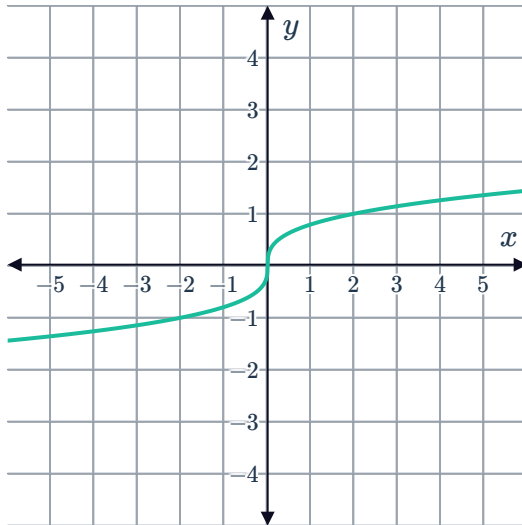
i



ii All real numbers.

iii All real numbers.

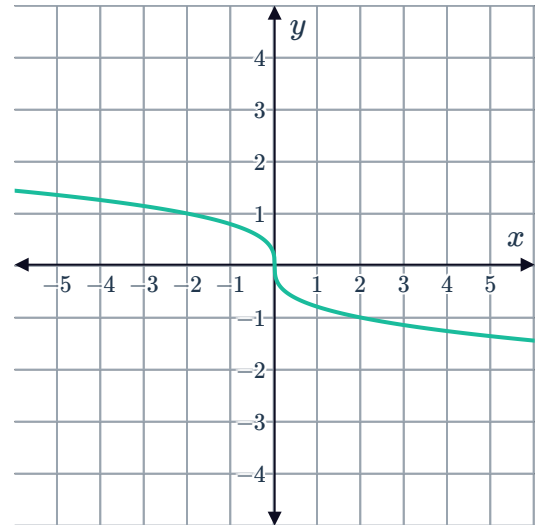
i
i



ii All real numbers.

iii All real numbers.

j
i



ii All real numbers.

iii All real numbers.

Characteristics of cube root functions

9 $y = 5\sqrt[3]{x}$ increases more rapidly than $y = \sqrt[3]{x}$.

10 All real x and all real y .

11



- a All real x .
- b All real y .
- c The functions $y = \sqrt[3]{x}$ and $y = \sqrt[3]{x - 4}$ increase at the same rate.